

Deep Integration of Export Destinations and Firm Innovation: Evidence from China ^{*}

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Abstract

How do firms in non-member countries adjust their innovation strategies when their export destinations engage in deep integration through preferential trade agreements (PTAs) membership? Using global PTA provisions and Chinese firm-level data, we construct the local exposure of an industry to the deep integration of its export destinations and find it negatively affects Chinese firms' innovation. Various negative trade effects along the intensive and extensive margins substantiate the competition in export destinations, in line with the Schumpeterian effect. We also find that the most productive firms innovate and profit more, whereas the least productive firms innovate less and incur losses, in response to the deep integration of their export destinations, supporting the escape-competition argument. This pattern is more pronounced in R&D intensive sectors. In addition, we show firms' innovation responses to the deep integration of export destinations vary by PTA provisions, China's PTA partnership, firms' patent types, upstreamness, and local IPR enforcement. The negative innovation effect, coupled with the deterioration of the terms of trade, suggests potential welfare losses for non-member countries.

Keywords: Innovation, Schumpeterian effect, Escape-competition effect, Preferential trade agreements, Deep integration

JEL codes: F13, F14, F15, F61, O31

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1 Introduction

The rapid expansion and deepening of preferential trade agreements (PTAs) have reshaped not only trade patterns but also the dynamics of global economic activity. Extensive literature examines the effects of PTAs on trade, innovation, and investment within member countries.¹ However, the literature has primarily focused on the trade effects for non-member countries when their export destinations engage in deep integration through PTA membership (Dai et al., 2014; Mattoo et al., 2022; Lee et al., 2023), with much less attention given to the third-country effects on other related economic variables such as innovation. Beyond trade, third-country firms' profitability, learning scope or access to knowledge could also be affected.² This, in turn, may alter their innovation incentives. Alternatively, these firms with different productivity levels may exhibit heterogeneous innovation strategies to navigate the intensified competition in the export markets. Understanding these complexities is essential for assessing the broader economic effects of PTAs.

In this paper, we fill this gap by studying how firms in third countries adjust their innovation behavior when their export destinations undergo deep integration through PTAs. Our analysis adopts China as a third country. Studying the innovation response of Chinese firms is important in its own right, as China is the largest developing economy and the second-largest economy in the world. Although China is a major global exporter, its PTA partners only account for around 6% of world GDP in 2013, and it has limited influence over the rules governing deep integration among other countries. As such, it is difficult for Chinese firms to manipulate the inter-connectedness and deep integration of export destinations. The exposure to the evolution of external deep integration of export destinations is thus exogenous to Chinese firms, allowing for a credible identification strategy.

We first construct the exposure of local industries in China to the deep integration of their export destinations (DI), by using the depth index of PTAs from the World Bank Deep Agreements Database (Hofmann et al., 2017), which defines the depth of a PTA as the number of provisions covered in a PTA. All country pairs within the same PTA are assigned to the same depth. With this dataset, the construction of DI involves two steps. First, we create a variable, PTA engagement, to capture

¹For example, Baier and Bergstrand (2007), Campi and Dueñas (2019), Baier et al. (2019), and Dutt (2020) study the trade effects of PTAs; Bustos (2011) studies the innovation effect of PTAs; and Baltagi et al. (2008) and Medvedev (2012) study investment effects of PTAs within member countries.

²Trade itself can serve as a conduit for learning and imitation. Using Chinese patent citation data, Liang et al. (2024) highlight the learning-by-exporting effect and demonstrate that there are actual knowledge transfers from destination markets to exporting firms, which help firms improve their innovation. Aghion et al. (2023) find that French firms disseminate knowledge to destination countries through exporting.

the extent to which each export destination is integrated with other markets (excluding China) through PTAs. PTA engagement for each destination is calculated as the sum of the pairwise PTA depth index across other markets. The exposure to PTA engagement of export destinations with other member countries could exert differential impact on Chinese local industries owing to local comparative advantages. DI is thus defined as a weighted sum of the PTA engagement of each export destination, with the weights equal to the initial local export share of the industry to these destinations.

To conduct empirical analysis, we first merge the patent filing data from the State Intellectual Property Office of China (SIPO), with the Annual Survey of Industrial Firms dataset (ASIF) that includes rich firm-level operation information. We focus on invention patent filings by manufacturing firms. Then we proceed to merge the matched data above with the DI index constructed. The final data allow us to study the effect of DI on Chinese firms' innovation, proxied by invention patent filings.

We find that the exposure of local industries to deeper integration of export destinations will result in Chinese firms filing for less invention patents over time. To isolate the innovation effect of DI, we control for concurrent MFN tariffs confronting each local industry, other innovation determinants such as local GDP and human capital, industry-year fixed effects that account for industrial policies and input-output linkages, and various firm-level characteristics. The results are robust to different subsamples and alternative measures of DI. Importantly, this paper emphasizes the inter-connectedness between export destinations. Thus, it is necessary to distinguish it from the own markets of export destinations confronting Chinese local industries. Our results are robust to controlling for the channel of destinations' own markets.

We explore potential mechanisms through which DI affects the innovation behavior of Chinese firms. Both the Schumpeterian effect and the escape-competition effect are at work. The Schumpeterian effect emphasizes the competition, which reduces the potential rents that a firm could capture from innovating and thus has a negative impact on firm innovation ([Schumpeter, 1942](#)). We substantiate the fierce competition confronting the Chinese firms in the export destinations by studying how DI affects the intensive and extensive margins of trade. Both export quantity and price are negatively affected, and are equally important channels in accounting for the trade diversion effect, a driving force of the negative innovation effect. Meanwhile, along the extensive margin, exporters export fewer products and to fewer countries. Nevertheless, we do not find significant effects on either

import quantity or import price. The decline of export price against unchanged import price implies a deterioration of terms of trade.

On the other hand, the escape-competition effect predicts that more productive firms will innovate more to escape competition from their rivals (Aghion et al., 2005). We find the least productive firms suffer profit losses and exhibit the greatest negative innovation response to DI. Interestingly, the negative effects are alleviated for more productive firms. And the most productive firms profit more and innovate more. We also find that the escape-competition effect is stronger in R&D intensive industries.

We further conduct several heterogeneous analysis to shed light on the underlying mechanisms. First, we examine the heterogeneous effects by PTA provisions. Preferential provisions reduce innovation primarily among low-productivity firms through trade diversion. By contrast, positive spillovers from non-discriminatory provisions (Mattoo et al., 2022; Lee et al., 2023) stimulate innovation among highly productive firms, which are better able to innovate to escape competition. Second, we study the role of China’s PTA partnership in mediating the effect of the deep integration of export destinations. We expect the competition induced by PTA links between China and its export destinations may drive greater innovation responses, particularly for firms at both ends of the productivity spectrum. We confirm a wider dispersion of innovation responses to the deep integration of China’s PTA partners, compared with that of China’s non-PTA partners. We also probe into the innovation type heterogeneity, and find that DI exerts a larger negative effect on filing for utility patents that benefit more from imitation. This is consistent with a learning story from export destinations through imitation: declines in exports limit the learning scope for utility patents (Liu et al., 2021). Interestingly, we find no correlation between design patent filings and DI. Finally, we find that the negative innovation response operates primarily through trade diversion in upstream segments of global value chains, namely intermediate goods. Moreover, the innovation effects of deep integration in export destinations critically depend on the strength of local intellectual property rights (IPR) enforcement.

This paper mainly contributes to two strands of the literature. First, this paper contributes to the literature on the effects of trade agreements on non-member countries. Starting with Viner (1950), the literature has extensively studied the trade creation within member countries and the third-country trade diversion effects of PTAs. Recent literature on the trade diversion effects includes Dai et al. (2014), Conconi et al. (2018), Maskus and Ridley (2021), Mattoo et al. (2022) and Lee et al. (2023). See Panagariya (2000) for a discussion on the conditions under which trade creation is associated with

a welfare gain and trade diversion with a welfare loss. [Mossay and Tabuchi \(2015\)](#) build a general equilibrium three-country model and show that a PTA lowers the welfare and the terms of trade in the third country. Nevertheless, all of these research papers focus on the trade effect. Only few papers study the third-country effects on other economic activities. Based on an incomplete contract model with endogenous contracting institution, [Luo and Hsu \(2024\)](#) find a negative third-country effect on institutional quality when member countries trade liberalize. We contribute to studying the third-country effect on innovation of PTAs, and find an adverse third-country effect. Our mechanism analysis also corroborates a negative effect on terms of trade. These results tend to imply a negative impact on welfare, supporting [Mossay and Tabuchi \(2015\)](#).

Second, this paper contributes to the rich literature on the innovation effect of trade. Export market opportunities motivate firms to innovate more ([Lileeva and Trefler, 2010](#); [Bustos, 2011](#); [Aw et al., 2011](#); [Liu and Ma, 2020](#); [Liang et al., 2024](#)), and there are heterogeneous effects across firms' productivity due to the market size effect and the induced competition effect, with more productive firms innovate more ([Aghion et al., 2022](#)). See also [Shu and Steinwender \(2019\)](#) and [Melitz and Redding \(2021\)](#) for surveys of the innovation effect of trade. We differ from the literature above in that we focus on the inter-connectedness of export destinations, instead of their own markets. We find that whereas there is a positive effect of destination's own markets, the inter-connectedness of export destinations exerts negative effects on innovation, with heterogeneous effects on innovation and profits across firms' productivity.

The rest of the paper is structured as follows. Section 2 introduces the data and the construction process of the deep integration of export destinations. Section 3 describes the empirical strategy and the estimation results, together with robustness checks. Section 4 explores potential mechanisms. We further conduct several heterogeneous analysis to shed light on the underlying mechanisms in section 5, and conclude in section 6.

2 Data and descriptive statistics

The empirical analysis is mainly based on four datasets. We first describe the datasets, and proceed to construct the measure of the deep integration of export destinations. Lastly, we explain the matching strategy and present the summary statistics.

2.1 Data

We obtain the patent filing data from State Intellectual Property Office of China (SIPO). The data contain detailed information on each patent filing since 1985, such as the date of filing, the name and address of each applicant, the name and the type of each patent, among others.

China recognizes three types of innovations—*invention*, *utility*, and *design*—with each governed by different requirements under Chinese Patent Law. Invention patents undergo a rigorous examination process that assesses their utility, novelty, and non-obviousness. The innovation must exhibit “prominently substantive characteristics and significant improvement” compared to existing technologies in order to be granted. In contrast, utility model and design patents are typically granted on a registration basis. Design patents primarily address a product’s appearance or shape, while utility patents still have to pass a novelty test and meet standards for practicality and functionality. We focus on invention patents in this paper.

The second data source is the Annual Survey of Industrial Firms (ASIF) dataset from 2000 to 2013, as maintained by the National Bureau of Statistics of China. It is the most comprehensive firm-level dataset in China, including all firms with annual sales above 5 million RMB (around US \$600,000 at the 2002 exchange rate). The threshold increased to 20 million RMB in 2011. In addition, it includes all state-owned firms with sales below 5 million RMB before 2007. ASIF contains the basic information for each surveyed firm, such as its identification number, location code, and industry affiliation, together with financial and operational information extracted from accounting statements, such as sales, employment, materials, fixed assets, age, capital, ownership, among others. We only focus on manufacturing firms in this paper.

Thirdly, we use the World Bank Deep Trade Agreements Database to capture the depth of trade agreements between country pairs ([Hofmann et al., 2017](#)). It maps the coverage of 52 selected policy areas in 279 PTAs notified at WTO signed between 1958 and 2015. These policy areas are further divided into two groups of 14 WTO+ and 38 WTO-X areas. WTO+ provisions in PTAs reconfirm existing commitments and, in some cases, provide for additional obligations. WTO-X provisions refer to policy areas that are not yet regulated by the WTO. Table B1 in the appendix lists the 52 provisions. As many of the 52 provisions go beyond trade issues, we focus on 18 core provisions defined by [Hofmann et al. \(2017\)](#): all WTO+ provisions and 4 WTO-X provisions (competition policy, investment, movement of capital and intellectual property rights). We will use these provisions to

construct the local exposure of an industry to the deep integration of its export destinations in the next subsection.

The fourth dataset is the Chinese Customs Data, which are monthly reports covering the universe of the Chinese export activities from 2000 to 2016. The key variables include export and import value, quantity, firm location, Harmonized System (HS) code at the 8-digit level, destination and firm name, among others. We aggregate the monthly transaction data to firm-destination-product-year level to avoid seasonality and irregular turnover patterns.³

We complement these datasets with other datasets, including MFN tariffs from the WITS (World Integrated Trade Solution) database, the Penn World Table, China’s City Statistical Yearbook, among others, to construct variables of interest and controls, which we will explain shortly.

2.2 Exposure to the deep integration of export destinations at the city-industry level

There has been a notable surge in the signing of PTAs across the globe. The number of PTAs grows to 279 in 2015 from 20 in 1990. Meanwhile, newer agreements are “deeper” in the sense that they generally cover a wider set of policy areas than older agreements (Mattoo et al., 2022). These trends can not only change trade flows among member countries and third countries, but also other related economic activities through trade.

In this paper, we study the third-country effect of the deep integration of export destinations on innovation, with China as the third country. Specifically, we construct the exposure to the deep integration of export destinations (DI) confronting local industries in China, and study how DI affects China’s innovation at the firm level.

Following Hofmann et al. (2017), we define the PTA depth as the number of core provisions covered in an agreement, and all country pairs within the same PTA have the same depth. More specifically, the depth of a PTA v is $\text{CoreDepth}_t^v = \sum_{i=1}^{18} \text{provision}_{it}^v$, where provision_{it}^v takes the value 1 if core provision i is covered in the PTA v at year t , and 0 otherwise.⁴ The PTA depth for country pair c and n is thus $\text{CoreDepth}_{cnt} = \text{CoreDepth}_t^v$, where country c and n are both in the trade agreement

³Product is defined at the HS 6 digit level, the most disaggregated codes that are consistently available for our entire study period.

⁴Hofmann et al. (2017) also assign values to provision_{it}^v based on the provision’s legal enforceability: 0 if the provision is not mentioned; 1 if the provision is mentioned, legally enforceable but explicitly excluded by dispute settlement; 2 if the provision is mentioned and legally enforceable.

v. We also consider the depth of a PTA based on the provision's legal enforceability or the sum of all the 52 provisions in the robustness checks.

Country c 's PTA engagement $Engagement_{ct}$ is defined as the sum of pairwise PTA depth between c and other countries (excluding China). Mathematically,

$$Engagement_{ct} = \sum_{n \neq China} CoreDepth_{cnt} / (N - 1),$$

where N is the number of total countries in the world. $Engagement_{ct}$ measures the simple average depth of PTAs that country c signed with other countries at t . We also consider weighted average depth in robustness checks, with different partners having weights according to initial trade flows.

The exposure to PTA engagement of export destinations could exert differential impacts on local economic activities owing to the local comparative advantages of different industries. We proceed to construct the average deep integration of export destination (DI) confronting a city-industry pair as follows:

$$DI_{rjt} = \sum_c Share_{rjc} Engagement_{ct}, \quad (1)$$

where r and j represent a city and a Chinese Industry Classification (CIC) 4-digit industry, respectively, and $Share_{rjc} = \frac{export_{rjc,2000}}{\sum_c export_{rjc,2000}}$ denotes the share of r 's exports to country c relative to its total exports in industry j to the world, such that $\sum_c Share_{rjc} = 1$. We fix the export share in the base year 2000 to reduce endogeneity issues (Autor et al., 2013; Borusyak et al., 2022). The distribution of DI is shown in Figure 1.

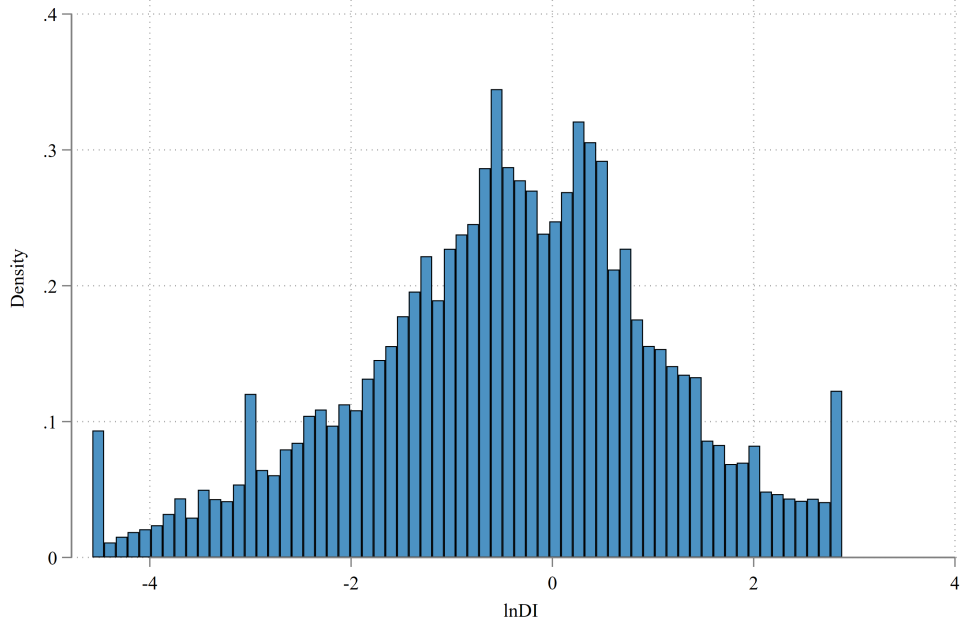


Figure 1: The distribution of $\ln DI$, the logarithm of DI (city-industry level), winsorized at the top and bottom 1%.

2.3 Data matching strategy and summary statistics

We first concord the CIC 4-digit industries with HS 6-digit product codes. To this end, we first convert all HS 6-digit product codes to those of the HS96 version, using the conversion tables provided by WITS that map between HS96, HS02, HS07, and HS12. Then, we map CIC 4-digit industries from the 2011 version to 2002 version. And finally, we map HS96 6-digit products to the CIC 4-digit industries, based on the concordance table from the National Bureau of Statistics of China.

Our matching strategy involves two steps. Firstly, we merge the patent filing dataset and ASIF by the firm name reported in both datasets (Liu et al., 2021). The matched data have an unbalanced panel of around 958,636 firms and a total of around 4,085,073 observations, with detailed patent filing information and firm characteristics between 2000-2013.⁵

Secondly, we merge the DI dataset obtained in subsection 2.2 with the matched dataset in the first step. The sample includes all firms within city-industry pairs that recorded exports in the year

⁵The matching quality resembles that of Liu et al. (2021). We have similar number of firms, observations and patent filings if we focus on the same period 1998-2005 as in Liu et al. (2021).

Table 1: Summary statistics

	obs	mean	sd	min	max
ln(DI)	2,701,295	-0.45	1.52	-4.57	2.88
Invention Patent Filings	2,730,185	0.21	12.54	0	5842
Utility Patent Filings	2,730,185	0.27	3.89	0	1924
Design Patent Filings	2,730,185	0.19	4.72	0	2207
ln(Firm Age)	2,728,288	3.15	0.31	2.56	4.20
ln(Employment)	2,709,739	4.84	1.10	2.08	7.79
ln(Capital-labor Ratio)	2,709,586	5.13	1.12	-0.75	12.04
Foreign share	2,730,185	0.13	0.34	0	1
Private share	2,730,185	0.78	0.41	0	1
Exporting Status	2,226,440	0.33	0.47	0	1

Note: The sample includes all firms within city-industry pairs that recorded exports in the year 2000. Data Sources: Authors' own calculation based on merged datasets from China National Bureau of Statistics, China's State Intellectual Property Office (SIPO), Annual Survey of Industrial Firms (ASIF), China's Customs, and the World Bank Deep Trade Agreements Database.

2000. Note around 56% of the city-industry pairs do not export in the sample in 2000, and hence firms in these pairs will not be taken into account in the analysis.⁶ Eventually, the final matched dataset consists of 634,043 firms and 2,730,185 observations. The sample used in our study covers the period from 2000 to 2013. Table 1 reports the summary statistics.

3 Firm-level analysis

3.1 Econometric specification

The empirical specification at the firm level is as follows:

$$Y_{f,rjt} = \alpha_0 + \alpha_1 \ln DI_{rjt} + \alpha_2 \mathbf{X}_{rjt} + \alpha_3 \text{firm_control}_{f,rjt} + \gamma_f + \mu_{jt} + \varepsilon_{f,rjt}, \quad (2)$$

where $Y_{f,rjt}$ refers to the innovation output made by firm f located in city r and operating in a CIC 4-digit industry j in year t , $\ln DI_{rjt}$ is the local exposure of industry j to the deep integration of its export destinations, and $\varepsilon_{f,rjt}$ is the error term. γ_f is the firm fixed effect so that the identifying

⁶Yu (2015) directly matched the ASIF dataset with the Customs dataset, and around 13.6% of manufacturing firms remain in the dataset. In contrast to his method, our matching in the second step keeps firms that do not export but are still affected by the DI shocks at the city-industry level.

variation is within each firm over time. It controls for all time-invariant determinants of firm-level innovation. μ_{jt} is the industry-year fixed effect so that we are comparing innovation among firms within the same industry in each year. It also controls for the input-output linkage such as industry-specific upstreamness and downstreamness, and industrial policies over time in China.

To isolate the effect of DI, we control for several time-varying factors $\mathbf{X}_{rjt}$ that may affect innovation. First, contemporary changes in MFN tariffs over time arising from accession to the WTO, can be correlated with the preferential trade liberalization mandated by PTAs. To rule out the possibility of capturing the effects of the MFN tariffs confronting local industries, we control for the weighted average tariffs tariff_{rjt} at the city-industry level, which is defined as $\text{tariff}_{rjt} = \sum_c \text{Share}_{rjc} \text{Tariff}_{cjt}$. Specifically, Tariff_{cjt} refers to the MFN tariff of industry j levied by country c , computed as the simple average of HS96 products within industry j ; the weight Share_{rjc} is the same as that defined previously in equation (1). We also control for other time-varying factors such as the local economic development proxied by local GDP per capita, and human capital by the number of local tertiary education institutions. These are key determinants of innovation. To avoid the potential “bad control” problem (Angrist and Pischke, 2009), we interact local GDP per capita and the number of local tertiary education institutions in the base year with a time trend. Therefore, $\mathbf{X}_{rjt} = \{\text{Tariff}_{rjt}; \text{GDP}_{r,2000} \times \text{trend}; \text{Human capital}_{r,2000} \times \text{trend}\}$.

We also control for several time-varying firm characteristics that may affect firm innovation, such as firm age, employment, capital-labor ratio, exporting status, and firm type, collected by the vector $\text{firm_control}_{f,rjt}$. To deal with the potential heteroskedasticity and serial autocorrelation, we cluster the standard errors at the city-industry level.

Our identifying assumption is that DI, the exposure of a local industry to the deep integration of its export destinations is exogenous to firms’ innovation, conditional on all the aforementioned controls. The reasons underlying the identifying assumption is as follows. Although China is a major global exporter, its PTA partners only account for around 6% of world GDP in 2013. As such, China has limited influence over the rules governing the deep integration among other countries. Hence, it is difficult for Chinese firms to manipulate the inter-connectedness and deep integration of export destinations, allowing for a credible identification strategy.

Given that there are many zero patent filings at the firm level, we apply the inverse hyperbolic sine (IHS) transformation to the number of patent filings in the baseline specification: $Y_{f,rjt} =$

$\log(y_{f,rjt} + \sqrt{1 + y_{f,rjt}^2})$, where $y_{f,rjt}$ is the number of firm f 's invention patent filings; see also [Atkin et al. \(2022\)](#) for similar transformations.

3.2 Main results

Table 2 shows the effect of the deep integration of export destinations on the patent filings of Chinese firms. We first adopt the IHS transformation in columns (1) to (4), followed by $\log(1 + y_{f,rjt})$ in column (5), and some alternative specifications from (6) to (8).

In column (1), we control for the firm fixed effect and the year fixed effect. The coefficient of DI is negative and statistically significant. This indicates that deeper integration of export destinations will result in less within-firm innovation over time. In column (2), we control for contemporary MFN tariffs confronting local industries, local economic development and local human capital and industry-year fixed effects. The results are similar. We further control for firm-level variables as in column (3), and the results remain stable. We use the regression setting in column (3) of Table 2 as the baseline specification. A one-standard-deviation increase in $\ln DI$ (1.52) reduces the number of patent filings by 0.3 percent. While this average effect appears modest at first glance, it masks substantial heterogeneity across firms with different productivity levels. As we shall see shortly in Figure 2, patent filings increase by 1.5 percent for the most productive firms (top TFP quintile), whereas they decline by 1.5 percent for the least productive firms (bottom TFP quintile). These estimated effects are comparable in magnitude to the impact of trade policy uncertainty on innovation reported by [Liu and Ma \(2020\)](#).

In column (4), we investigate whether there is any anticipatory effect by also controlling for DI one year before the focal year, $\ln DI_{t-1}$, and that one year after the focal year, $\ln DI_{t+1}$ ([Aghion et al., 2022](#); [Bai et al., 2024](#)). A significant coefficient of $\ln DI_{t+1}$ would indicate that future DI affects current patent filings, suggesting an anticipatory effect at work, and thus casting doubt on the finding that DI results in less within-firm innovation. Reassuringly, the coefficient of $\ln DI_{t+1}$ is statistically insignificant. On the other hand, the coefficient of $\ln DI_{t-1}$ is statistically significant, suggesting a lagged effect.

In column (5), we apply the log transformation to the number of patent filings after adding 1, and the coefficient of DI remains significantly negative. As pointed out by [Chen and Roth \(2024\)](#), using the IHS or $\log(1 + y_{f,rjt})$ can have interpretational issues when handling variables with zero values,

Table 2: Main results

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Arcsinh(patents)				log(1+patents)	Indicator	Entry	CDF
lnDI	-0.0017** (0.0006)	-0.0018*** (0.0006)	-0.0019*** (0.0006)	-0.0012 (0.0009)	-0.0015*** (0.0005)	-0.0013*** (0.0003)	-0.0009*** (0.0003)	-1.96e-05*** (6.49e-06)
lnDI _{t-1}				-0.0018** (0.0009)				
lnDI _{t+1}				0.0013 (0.0010)				
Tariff		0.0005 (0.0010)	0.0007 (0.0010)	0.0007 (0.0010)	0.0005 (0.0008)	0.0002 (0.0005)	0.0002 (0.0005)	0.0000 (0.0000)
GDP _{r,2000} x trend		0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	-0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
Human capital _{r,2000} x trend		0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0000** (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
Firm control			✓	✓	✓	✓	✓	✓
Industry-year FE		✓	✓	✓	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓
N	2514576	2454623	2431176	2345554	2431176	2431176	1833298	2431176
R ²	0.517	0.529	0.528	0.534	0.535	0.452	0.331	0.490

Note: Standard errors are clustered at the city-industry level. The indicator in Column 6 takes the value 1 if the firm has positive invention patents in year t , and 0 otherwise. Entry is defined as first-time filing for invention patents. CDF is the cumulative distribution function of invention patent filings. We control for firms' age, employment, capital labor ratio, firm type and exporting status. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

due to unit dependence of such transformations. To resolve the issue of unit dependence, we follow [Chen and Roth \(2024\)](#) and first study the effect of DI at the extensive margin, using the patent filing dummy or entry dummy as the dependent variable. Specifically, Indicator _{f,rjt} takes the value 1 if the firm has at least one patent filing in year t , and 0 otherwise. The results are shown in column (6). The coefficient is -0.0013, suggesting that a one standard deviation increase in lnDI (1.52) reduces the fraction of patent filings by 0.2%. This is a moderate effect compared to the 3% share of firms with at least one patent filing.⁷ In column (7), we construct a dummy variable Entry to indicate whether a firm starts filing for patents. If a firm did not file for a patent in $t - 1$ but did file in t , then Entry = 1; if the firm did not file for a patent in either $t - 1$ or t , then Entry = 0; otherwise, it takes missing values. The effect is again negative and statistically significant.

As recommended by [Chen and Roth \(2024\)](#) and [Delius and Sterck \(2024\)](#), we also use the cumulative distribution function (CDF) of patent filings as the dependent variable to avoid the issue of unit dependence. This transformation represents the rank (i.e., percentile) of a firm within the overall distribution. We report the estimation results in column (8). The coefficient of DI remains negative and statistically significant.

⁷The mean value of Indicator _{f,rjt} is 3%.

We also use the Poisson model to deal with the zero patent filings. In addition, we run a regression with baseline specification, but focus only on positive patent filings that is logged as the dependent variable. Neither result is significant.⁸ Taking together, these results in Table 2 suggest that deeper integration of export destinations will result in firms innovating less over time, and the effect of DI mainly works along the extensive margin.

3.3 Robustness checks

In this subsection, we take into account additional controls and alternative measures of DI. The results are shown in Table 3, and are robust to these considerations. Column (1) shows the baseline result for comparison, as in column (3) of Table 2.

Market size — Recall DI reflects the average deep integration of export destinations with other countries. Therefore, it offers greater market potential for Chinese exporters. Nevertheless, we need to distinguish the interconnectedness of export destinations from the own markets of export destinations confronting Chinese local industries.

We construct the average market size confronting a local industry as follows:

$$\text{Market Size}_{rjt} = \sum_c \text{Share}_{rjc} \text{GDP}_{ct}, \quad (3)$$

where GDP_{ct} is the GDP of export destination c in year t , and Share_{rjc} is the same as in equation (1).

We control for the market size channel, as shown in column (2) of Table 3. As expected, the coefficient of the market size is positive. After teasing out the effect of destinations' own market size, the innovation effect of DI slightly worsens. Importantly, the innovation effect of DI is comparable to that of the market size, though with opposite signs.

Threshold effect — Note that the ASIF dataset includes firms with designated threshold, which increases from 5 million RMB to 20 million RMB in 2011. To rule out the possibility of our results being affected by the change in the firm size threshold, we focus on the subsample before year 2011, and re-estimate the effect of DI. The estimation result is similar, as shown in column (3) of Table 3.

⁸Estimation results are available upon request.

Table 3: Robustness checks

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Baseline	Market size	Year ≤ 2010	Year ≥ 2002	Arcsinh(Patents)		Alternative DI measures			
lnDI	-0.0019*** (0.0006)	-0.0021*** (0.0006)	-0.0017*** (0.0005)	-0.0020*** (0.0007)						
Market size		0.0013* (0.0007)								
lnDI(weighted PTA engagement)					-0.0014** (0.0007)					
lnDI(LE)						-0.0018*** (0.0006)				
lnDI(WTO+)							-0.0019*** (0.0006)			
lnDI(WTO X4)								-0.0016** (0.0006)		
lnDI(Total)									-0.0019*** (0.0006)	
lnDI(DESTA)										-0.0020*** (0.0007)
Firm control	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$\mathbf{X}_{rjt}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Industry-year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
N	2431176	2430923	2040166	2254633	2431176	2431176	2431176	2400149	2431176	2438370
R^2	0.528	0.528	0.533	0.540	0.528	0.528	0.528	0.530	0.528	0.528

Note: Standard errors are clustered at the city-industry level. We control for firms' age, employment, capital labor ratio, firm type and exporting status. $\mathbf{X}_{rjt}$ includes tariffs, local GDP trend and human capital trend. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

WTO accession effect — The sample period covers year 2000 to 2013, and thus the estimation results may capture the effect of China's accession to the WTO in December 2001. To exclude this possibility, besides controlling for the MFN tariffs, we also focus on the subsample from year 2002 onwards, and re-estimate the effect of DI. The estimation result is comparable to the baseline estimate, as shown in column (4) of Table 3.

Alternative measures of DI — As mentioned before, we compute each country's PTA engagement using a simple average, treating the importance of member countries equal. We construct an alternative measure of PTA engagement by assigning different degrees of influence to member countries according to their initial trade share. Specifically, $Engagement'_{ct} = \sum_{n \neq China} \frac{Export_{cnt_0}}{Export_{ct_0}} CoreDepth_{cnt}$, where $Export_{cnt_0}$ is the exports of country c to n in initial period t_0 , and $Export_{ct_0} = \sum_{n \neq China} Export_{cnt_0}$ is the total exports of country c to all countries excluding China. We re-construct DI using the weighted PTA engagement, and show the regression results in column (5). The magnitude of the coefficient is comparable with the baseline estimate. We also construct DI based on the legal enforceability of provisions and the results are presented in column (6). In addition, we use the 14 WTO+ provisions, the 4 core WTO-X provisions, and all the 52 provisions to construct DI, and the results are shown in column (7), (8) and (9), respectively. Finally, we construct DI by using the Design

of Trade Agreements Database (DESTA) database in column (10).⁹ These estimates are stable and robust across different specifications.

3.4 Additional evidence at the city-industry level

Note the dependent variable is at the firm level whereas the regressor of interest is at the city-industry level. The estimate we present so far is thus the local industrial average effect. We show additional evidence by aggregating the patent filings to the city-industry level and running regression at the city-industry level to check the robustness of our main finding.

Our econometric model at the city-industry level is specified as follows:

$$Y_{rjt} = \beta_0 + \beta_1 \ln DI_{rjt} + \beta_2 \mathbf{X}_{rjt} + \gamma_{rj} + \mu_{jt} + \varepsilon_{rjt}. \quad (4)$$

where Y_{rjt} is the IHS transformation of total invention patent filings for industry j in city r in year t . We control for the city-industry fixed effect γ_{rj} so that the identifying variation is within each city-industry pair over time, and γ_{rj} accounts for all time-invariant determinants of innovation at the city-industry level, such as local comparative advantages. As before, $\mathbf{X}_{rjt} = \{\text{Tariff}_{rjt}; \text{GDP}_{r,2000} \times \text{trend}; \text{Human capital}_{r,2000} \times \text{trend}\}$, μ_{jt} is the industry-year fixed effects. We cluster the standard errors at the city-industry level.

Table 4 reports the estimation results. We only control for city-industry fixed effects, and year fixed effects in column (1). The coefficient of DI is negative and statistically significant, indicating that the deeper integration of export destinations will negatively affect China’s innovation at the city-industry level over time. Column (2) controls for $\mathbf{X}_{rjt}$, and the coefficient of DI remains negative and statistically significant. In column (3), we compute DI based on the sum of all the 52 provisions, and the results are similar.

To measure innovation at the city-industry level, [Kou and Liu \(2017\)](#) propose an alternative innovation index at the city-industry level by aggregating patent values through a patent renewal model.¹⁰

⁹The Design of Trade Agreements Database (DESTA) looks in more detail at the content and depth of specific provisions in PTAs ([Dür et al., 2014](#)). Specifically, the coding covers a number of aspects for each area ranging from the simple inclusion of a chapter in the PTA to details on national treatment or harmonization of policies. Nevertheless, the coverage is only limited to 10 policy areas.

¹⁰The data are available from the “China Urban and Industrial Innovation Report 2017”, compiled by the Industrial Development Research Center in Fudan University.

Table 4: Innovation effect of DI at the city-industry level

	(1)	(2)	(3)	(4)
	Arcsinh(patents)			
lnDI	-0.0105*** (0.00134)	-0.0115*** (0.00147)		-0.00216*** (0.00027)
lnDI (Total)			-0.0135*** (0.00139)	
Tariff		-0.0105*** (0.00187)	-0.0109*** (0.00186)	0.00117*** (0.00032)
GDP _{<i>r</i>,2000} x trend		0.0103*** (0.000331)	0.0102*** (0.000331)	0.00379*** (0.00018)
Human capital _{<i>r</i>,2000} x trend		0.00172*** (0.000167)	0.00170*** (0.000167)	0.00211*** (0.00009)
City-industry FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Industry-year FE		✓	✓	✓
<i>N</i>	591281	518651	518651	542558
<i>R</i> ²	0.389	0.474	0.474	0.680

Note: Standard errors are clustered at the city-industry level. We aggregate quantity of invention patents to the city-industry level in Column (1) to (3). Column (4) uses monetary patent values computed by [Kou and Liu \(2017\)](#). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

This alleviates concerns of quality and value heterogeneity across patents to some extent, compared with directly using the total number of filed patents at the city-industry level. The distribution of the innovation index is shown in Figure A1. Our findings hold when using this alternative innovation index, as shown in column (4). Overall, we find that the deep integration of export destinations has a negative effect on China's innovation at the city-industry level.

4 Mechanisms

In this section, we explore potential mechanisms through which the deep integration of export destinations with other countries affects the innovation behavior of Chinese firms. One natural mechanism is the Schumpeterian effect ([Schumpeter, 1942](#)): the deep integration of export destinations intensifies the competition for Chinese firms and reduces the potential rents that a firm could capture from innovating. This in turn has a negative impact on firm innovation. To substantiate the fierce competition confronting the Chinese firms, we will first focus on how DI impacts the intensive margin

of trade. Specifically, we examine the classical trade diversion channel and its decomposition into the price effect and the quantity effect. Then we study the effect of DI at the extensive margin of the trade.

Equally important, a counteracting force — the escape-competition effect could be prevalent. When firms are locked in neck-and-neck competition in the international market, more productive firms will innovate more to escape competition from their rivals (Aghion et al., 2005). Hence, we also study how firms with different initial productivity levels react to DI.

4.1 The Schumpeterian effect

We use detailed Customs data at the firm-product-destination level to study the effect of DI on exports and imports, with products defined at the HS 6-digit level. This data offers the additional advantage of enabling the decomposition of sales into price and quantity, thereby providing insights into the underlying mechanisms.

The empirical specification is as follows:

$$Y_{f,p,rjt} = \theta_0 + \theta_1 \ln DI_{rjt} + \theta_2 \mathbf{X}_{rjt} + \gamma_{ft} + \mu_{jt} + \varepsilon_{f,rjt}, \quad (5)$$

where $Y_{f,p,rjt}$ refers to the logged exports or imports of product p of firm f located in city r and industry j in year t . γ_{ft} is the firm-year fixed effect so that the firm-level time-varying controls are absorbed, thanks to the more granular data. The rest variables are the same as in equation (2).

Table 5 presents the trade effect of the deep integration of export destinations at the firm-product level. Column (1) to column (3) report the effect of DI on exports, and column (4) to column (6) on imports. Column (1) shows that DI results in trade diversion for Chinese exporters. Both quantity and price of exported products are negatively affected, as shown in column (2) and (3). The magnitude of the two estimates are similar, indicating that both quantity and price are important channels in accounting for the trade diversion effect. In particular, the negative effect on price corroborates the fierce competition in the destination markets. Intuitively, the deep integration of export destinations, due to preferential provisions covered in PTAs, exempts members from various trade barriers and regulatory requirements. Therefore, member countries can expand sales within PTAs, driving down prices and diverting trade away from Chinese firms. On the other hand, non-discriminatory provisions

Table 5: Trade effect of DI at the firm-product level: Intensive margin

	(1)	(2)	(3)	(4)	(5)	(6)
	Value	Export Quantity	Price	Value	Import Quantity	Price
lnDI	-0.0278*** (0.0059)	-0.0162** (0.0080)	-0.0116** (0.0055)	-0.0096* (0.0052)	-0.0010 (0.0085)	-0.0087 (0.0059)
$\mathbf{X}_{rjt}$	✓	✓	✓	✓	✓	✓
Firm-year FE	✓	✓	✓	✓	✓	✓
Industry-year FE	✓	✓	✓	✓	✓	✓
N	25,741,546	25,741,546	25,741,546	16,800,261	16,800,261	16,800,261
R^2	0.3038	0.3792	0.5666	0.3047	0.4489	0.5959

Note: Standard errors are clustered at the city-industry level. $\mathbf{X}_{rjt}$ includes tariffs, local GDP trend and human capital trend. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

towards third countries can also reduce trade costs and non-trade barriers. This further lower prices in destinations markets (Mattoo et al., 2022). Overall, competition becomes fiercer in the destination markets. Column (4) shows imports are not much affected by DI, though import values slightly decrease. Neither import quantity nor price is significantly affected, as shown in column (5) or (6). Notably, the decline of export price versus unchanged import price implies a deterioration of the terms of trade, reminiscent of the theoretical finding that PTAs lower the welfare and the terms of trade in the third country (Mossay and Tabuchi, 2015).

Besides the effect on the intensive margin, we also investigate the effect of DI on the extensive margins of exports and imports. In particular, we use as dependent variables the number of HS6 products each firm exports and imports, and the number of destinations each firm exports to or imports from. The regression specification is thus at the firm level. Table 6 shows the effect of DI at these extensive margins. Column (1) and (2) show that exporters would export fewer products and to fewer countries in response to deeper integration of export destinations. Nevertheless, importers are not much affected on both margins, as shown in column (3) and (4).

Table 5 and 6 substantiate the fierce competition in the export destinations, not only along the intensive margins (exporters cut prices and quantities), but also along the extensive margins (exporters shrink product lines or exit part of export markets), which in turn discourage firms from innovating, consistent with the Schumpeterian effect (Schumpeter, 1942).

Table 6: Trade effect at the firm level: Extensive margin

	(1)	(2)	(3)	(4)
	Export		Import	
	# Products	# Countries	# Products	# Countries
lnDI	-0.2736*	-0.1086***	-0.0650	-0.0130*
	(0.1409)	(0.0340)	(0.0423)	(0.0069)
$\mathbf{X}_{rjt}$	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓
Industry-year FE	✓	✓	✓	✓
N	2,029,088	2,029,088	1,271,372	1,271,372
R^2	0.7586	0.7586	0.8023	0.7233

Note: Standard errors are clustered at the city-industry level. $\mathbf{X}_{rjt}$ includes tariffs, local GDP trend and human capital trend. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.2 The escape-competition effect

The deep integration of export destinations results in intensified competition. Such competition can make less productive firms profit less, and thus hamper innovation. On the other hand, it can also motivate more productive firms to innovate more in order to escape competition from other firms in the export markets.

We explore whether the innovation response varies across firms based on their initial productivity. Specifically, we adopt the method of [Levinsohn and Petrin \(2003\)](#) to estimate firms' total factor productivity (TFP), and group firms into quintiles based on their initial TFP within each industry. Based on the baseline specification as in equation (2), we interact DI with firms' initial productivity quintiles. Specifically, we use the following specification:

$$Y_{f,rjt} = \alpha_0 + \sum_g \alpha_1^g \text{Quintile}_f^g \times \ln \text{DI}_{rjt} + \alpha_2 \mathbf{X}_{rjt} + \alpha_3 \text{firm_control}_{f,rjt} + \gamma_f + \mu_{jt} + \varepsilon_{f,rjt}, \quad (6)$$

where Quintile_f^g indicates the corresponding quintile g that firm f 's initial TFP belongs to within its own industry, with integer g ranging from 1 to 5.

Figure 2 demonstrates the heterogeneous innovation effects by TFP. Firms in the lowest TFP quintile exhibit the greatest negative innovation response to DI. Interestingly, the negative effect on innovation is alleviated as productivity increases, with firms in the highest TFP quintile showing

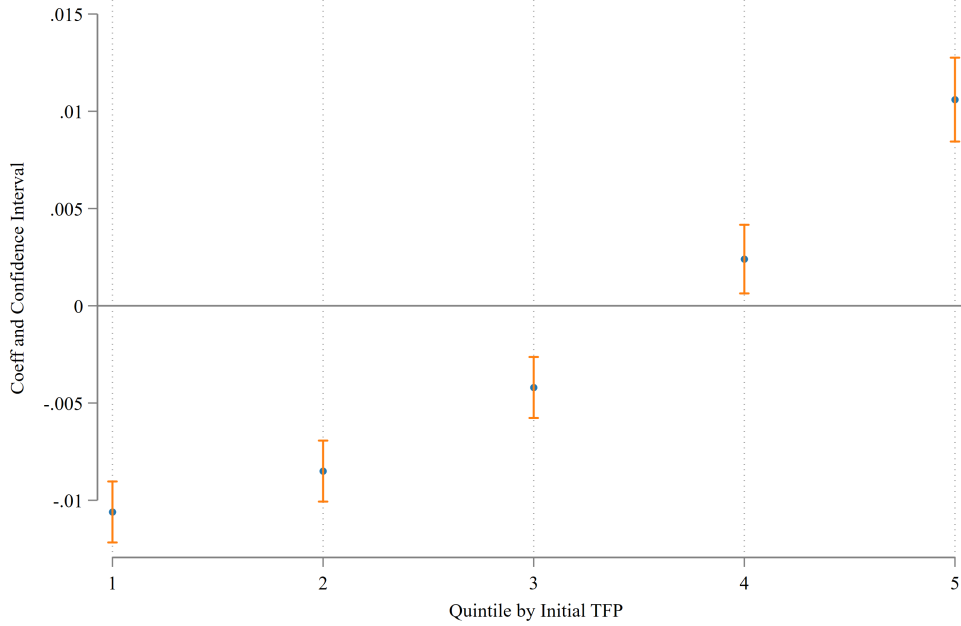


Figure 2: Heterogeneous innovation effects by TFP. The plot shows the coefficient α_1^g of regression (6) for each TFP quintile g and the corresponding 95% confidence interval, where g ranges from 1 to 5. The dependent variable is firm f 's invention patents.

the strongest positive innovation response. Figure 3 provides evidence that firms in the lower TFP quintile incur losses, whereas those in the higher TFP quintile profit more in response to the deep integration of export destinations. The heterogeneous results for innovation and profits indicate the presence of both the Schumpeterian effect and the escape-competition effect.¹¹

We also find that the escape-competition effect is stronger in R&D intensive sectors.¹² As illustrated in Figure A2, firms in sectors with high R&D intensity have more dispersed response of innovation to the deep integration of export destinations. This pattern aligns with their profit responses, as shown in Figure A3: compared to firms in low R&D intensity sectors, the most productive firms in high R&D intensity sectors achieve the highest profit gains and the strongest innovation responses. In contrast, the least productive firms in high R&D intensity sectors face the greatest profit

¹¹Even though there are heterogeneous profit effects in response to DI by TFP, the average profit effect is not statistically significant.

¹²Feng et al. (2016) calculate the R&D intensity measures as the ratio of industry R&D costs relative to industry output value for each 4-digit CIC industry. They define the sectoral R&D intensity as the mean value of R&D intensity measures for the 4-digit CIC industries within each 2-digit sector. We take these values and categorize manufacturing sectors into two types: low and high R&D intensity sectors.

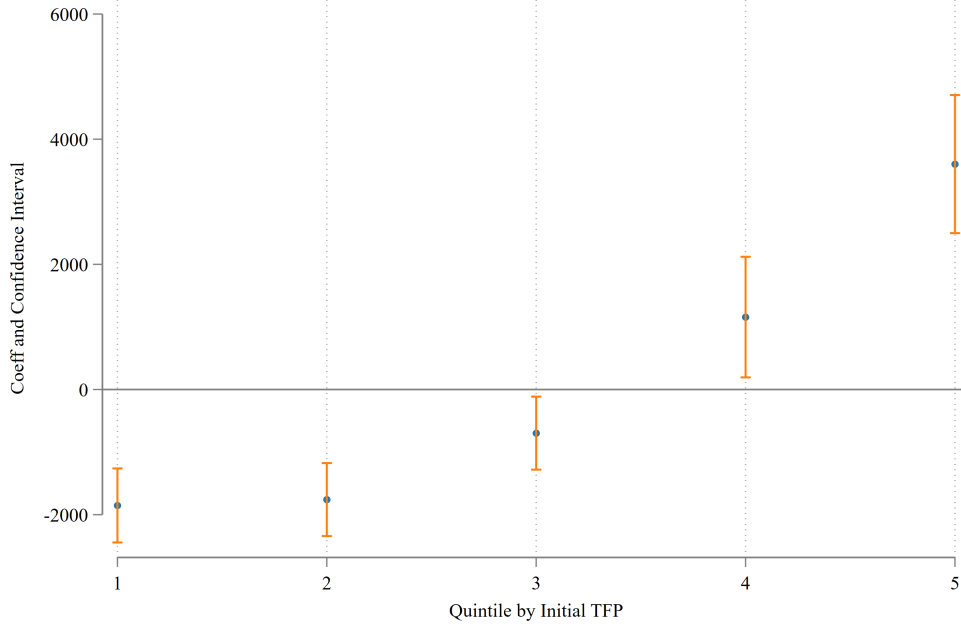


Figure 3: Heterogeneous profit effects by TFP. The plot shows the coefficient α_1^g of regression (6) for each TFP quintile g and the corresponding 95% confidence interval, where g ranges from 1 to 5. The dependent variable is firm f 's profit level.

losses and the weakest innovation performance.

5 Further heterogeneous effects

5.1 Heterogeneity by PTA provisions

We explore the heterogeneous effects by PTA provisions. [Baldwin and Low \(2009\)](#) classify PTA provisions in two groups: preferential provisions and non-discriminatory (i.e., “most-favoured nation” or MFN) provisions. The former provisions are these that apply only to the countries that signed the PTA; whereas the latter provisions are applied on an MFN basis. Preferential provisions are more likely to accrue benefits exclusively to the member countries, inflicting a further competitive disadvantage on third countries. The resulting trade diversion effect can lead to a negative innovation effect. On the other hand, MFN provisions can generate positive spillovers to third countries ([Mattoo et al., 2022](#); [Lee et al., 2023](#)), which can lead to a positive innovation effect.

We classify the core provisions in PTAs into preferential provisions denoted as {Pref}, and MFN provisions, denoted as {MFN}, based on Table B1. Then we decompose the core depth of each PTA into two types: $\text{CoreDepth}_t^{\text{Pref}} = \sum_{i \in \text{Pref}} \text{provision}_{it}^v$, and $\text{CoreDepth}_t^{\text{MFN}} = \sum_{i \in \text{MFN}} \text{provision}_{it}^v$. Following Subsection 2.2, we can similarly define $\text{DI}_{rjt}^{\text{Pref}}$ based on preferential provisions and $\text{DI}_{rjt}^{\text{MFN}}$ based on MFN provisions.

To study the heterogeneous effect of provisions in the presence of escape-competition effect, we interact $\text{DI}_{rjt}^{\text{Pref}}$ and $\text{DI}_{rjt}^{\text{MFN}}$ with firms' initial productivity quintiles, respectively. Specifically, we use the following specification:

$$Y_{f,rjt} = \alpha_0 + \sum_g \alpha_1^g \text{Quintile}_f^g \times \ln \text{DI}_{rjt}^{\text{Pref}} + \sum_g \alpha_2^g \text{Quintile}_f^g \times \ln \text{DI}_{rjt}^{\text{MFN}} + \alpha_3 \mathbf{X}_{rjt} + \alpha_4 \text{firm_control}_{f,rjt} + \gamma_f + \mu_{jt} + \varepsilon_{f,rjt}. \quad (7)$$

The innovation responses along the initial productivity dimension to $\text{DI}_{rjt}^{\text{Pref}}$ and $\text{DI}_{rjt}^{\text{MFN}}$ are plotted in Figure 4. The left panel shows that negative innovation effects are concentrated among the least productive firms, while more productive firms exhibit negative but statistically insignificant responses. This pattern reflects the interaction of two opposing forces. On the one hand, more productive firms have stronger incentives to innovate in order to escape competition. On the other hand, preferential provisions generate trade diversion for the third country China, which dampens innovation incentives of Chinese firms. For highly productive firms, these forces largely offset each other, whereas for low-productivity firms the weaker escape-competition effect amplifies the trade-diversion channel, resulting in negative innovation responses.

The right panel in Figure 4 shows a significantly positive innovation response among the most productive firms. The large magnitude of this effect is consistent with positive spillovers from MFN provisions to third countries—such as China—reinforced by strong escape-competition incentives at the top of the productivity distribution. In contrast, the least productive firms experience negative innovation effects, indicating that the potential spillovers from MFN provisions are dominated by their limited ability to innovate in response to intensified competition.¹³

¹³The negative innovation effects for the least productive firms is consistent with [Chen and Mattoo \(2008\)](#), who argue that although MFN provisions may generate scale economies, non-member countries can experience export losses when restrictive rules of origin and the raised stringency of standards exclude them from these benefits.

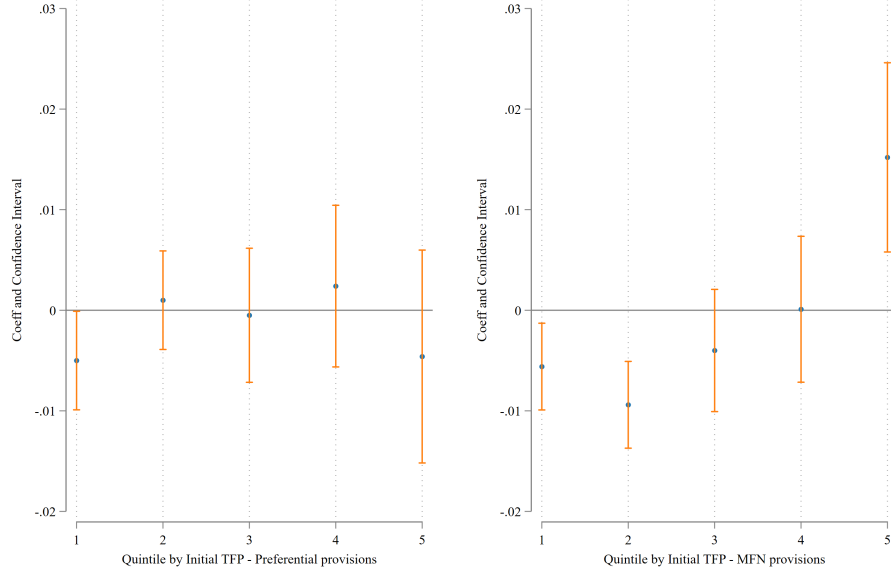


Figure 4: Heterogeneous innovation effects by PTA provisions. DI_{rjt}^{Pref} and DI_{rjt}^{MFN} are constructed based on preferential and MFN provisions, respectively. We interact DI_{rjt}^{Pref} and DI_{rjt}^{MFN} with firms' initial productivity quintiles, respectively, as in equation (7). The plots show the coefficient α_1^g and α_2^g for each TFP quintile g , respectively, with the corresponding 95% confidence interval, where g ranges from 1 to 5. The dependent variable is firm f 's invention patents.

5.2 Heterogeneity by China's PTA partner vs non-partner countries

In this subsection, we analyze how China's PTA membership mediates the innovation effect of the deep integration of export destinations. Intuitively, destinations that are China's PTA partners (as represented by country B in Figure 5), on average, tend to be more competitive for Chinese firms, compared to those that are not China's PTA partners (as represented by country A in Figure 5), conditional on the same deep integration with other countries. The competition induced by PTA links between China and its export destinations may amplify innovation responses, particularly for firms at both ends of the productivity spectrum. Specifically, we expect that more productive firms will innovate more and less productive firms will innovate less, in response to DI when exporting to China's PTA partners.

We first classify export destinations into China's PTA partners, denoted as $\{P\}$, and non-PTA partner countries, denoted as $\{NP\}$, based on whether China has ever signed PTAs with these coun-

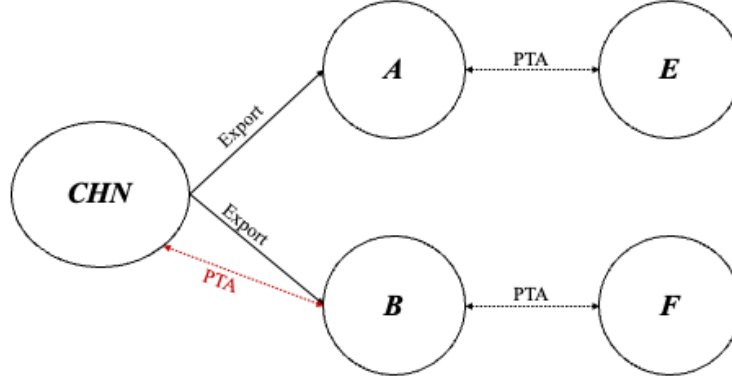


Figure 5: China's PTA partner country B and non-partner country A.

tries. Then we compute the deep integration of the two types of export destinations, respectively. Specifically, the deep integration of China's PTA partners DI_{rjt}^p is defined as follows:

$$DI_{rjt}^p = \sum_{c \in \{P\}} Share_{rjc} Engagement_{ct};$$

And the deep integration of China's non-PTA partners DI_{rjt}^{np} is defined as follows:

$$DI_{rjt}^{np} = \sum_{c \in \{NP\}} Share_{rjc} Engagement_{ct}.$$

We interact DI_{rjt}^p and DI_{rjt}^{np} with firms' initial productivity quintiles, respectively. Specifically, we use the following specification:

$$Y_{f,rjt} = \alpha_0 + \sum_g \alpha_1^g Quintile_f^g \ln DI_{rjt}^p + \sum_g \alpha_2^g Quintile_f^g \ln DI_{rjt}^{np} + \alpha_3 \mathbf{X}_{rjt} + \alpha_4 \text{firm_control}_{f,rjt} + \gamma_f + \mu_{jt} + \varepsilon_{f,rjt}. \quad (8)$$

The innovation responses along the initial productivity dimension to DI_{rjt}^p and DI_{rjt}^{np} are plotted in Figure 6. As expected, we find a stronger escape-competition effect for firms that export to China's PTA partners (right panel), i.e., a wider dispersion of innovation responses to the deep integration of China's PTA partners, compared with that of China's non-PTA partners (left panel).

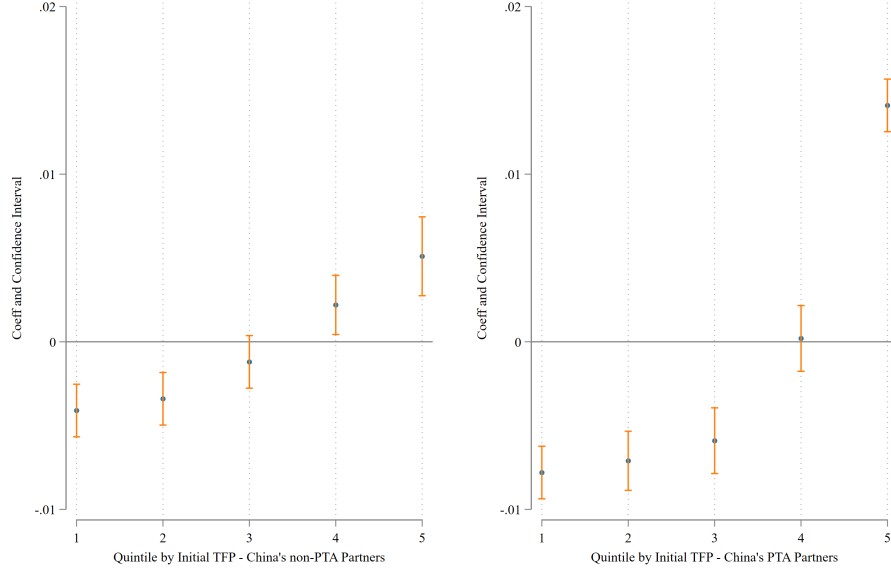


Figure 6: Heterogeneous innovation effects by China's PTA partnership. DI_{rjt}^p and DI_{rjt}^{np} are constructed based on China's PTA partners and non-partners, respectively. We interact DI_{rjt}^p and DI_{rjt}^{np} with firms' initial productivity quintiles, respectively, as in equation (8). The plots show the coefficient α_1^g and α_2^g for each TFP quintile g , respectively, with the corresponding 95% confidence interval, where g ranges from 1 to 5. The dependent variable is firm f 's invention patents. The dependent variable is firm f 's invention patents.

5.3 Innovation type heterogeneity

We study whether the impact of the deep integration of export destinations on Chinese firms' innovation varies by patent types: invention, utility, and design patents.

Table 7 reports the heterogeneous impacts. Column (1) shows the effect on invention innovation for comparison. The negative effect on utility innovation is even greater, as shown in column (2). This result is consistent with a learning story from export destinations (Liang et al., 2024). Liu et al. (2021) highlight that utility patents benefit more from positive spillover effects through imitation. Declines in exports lead to less learning scope for firms to engage in utility innovation, which results in a greater negative effect on filing for utility patents. Column (3) indicates no correlation between design patent filings and the deep integration of export destinations. This suggests a limited role of design patents in competing for international markets, especially if the competition is more related to technological advances or cost-efficiency rather than aesthetics (appearance or shape). As a result,

Table 7: Heterogeneity results by patent types

	(1)	(2)	(3)
	Patent types		
	Invention	Utility	Design
DI	-0.0019*** (0.0006)	-0.0037*** (0.0007)	-0.0007 (0.0005)
Firm control	✓	✓	✓
$\mathbf{X}_{rjt}$	✓	✓	✓
Firm FE	✓	✓	✓
Industry-year FE	✓	✓	✓
N	2431176	2431176	2431176
R^2	0.528	0.511	0.441

Note: Standard errors are clustered at the city-industry level. We control for firms' age, employment, capital labor ratio, firm type and exporting status. $\mathbf{X}_{rjt}$ includes tariffs, local GDP trend and human capital trend. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 8: Heterogeneous effects by industry upstreamness and IPR protection

	(1)	(2)	(3)	(4)
	Downstreamness	Upstreamness	Low IPR	High IPR
DI	-0.0013* (0.0008)	-0.0033** (0.0013)	-0.0021*** (0.0007)	0.0003 (0.0010)
Firm control	✓	✓	✓	✓
$\mathbf{X}_{rjt}$	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓
Industry-year FE	✓	✓	✓	✓
N	1261188	1156137	1073201	1348081
R^2	0.501	0.569	0.536	0.527

Note: Standard errors are clustered at the city-industry level. We control for firms' age, employment, capital labor ratio, firm type and exporting status. $\mathbf{X}_{rjt}$ includes tariffs, local GDP trend and human capital trend. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

firms might maintain a steady rate of design patent applications, regardless of the level of external competition.

5.4 Other dimensions

Upstreamness We examine heterogeneity along the dimension of firms’ industry upstreamness. We measure upstreamness using the industry-level index proposed by [Antràs et al. \(2012\)](#), which is based on input–output tables and captures an industry’s average distance from final use. We split industries into upstream and downstream groups using the median value of this index.

Columns (1) and (2) of Table 8 show that firms operating in more upstream industries experience significantly more negative innovation effects. By contrast, firms in downstream industries exhibit weaker and less pronounced effects. This pattern suggests that the negative innovation response driven by trade diversion operates primarily through upstream segments of global value chains, namely intermediate goods. This finding is consistent with [Conconi et al. \(2018\)](#), who show that rules of origin in PTAs such as NAFTA substantially reduced imports of intermediate inputs from third countries relative to member countries.

Local IPR Enforcement We further examine heterogeneity along the dimension of firms’ local intellectual property rights (IPR) enforcement. Local IPR institutions shape firms’ incentives to innovate by affecting the appropriability of innovation returns and the capability of escaping competition. We measure provincial IPR enforcement using the Intellectual Property Protection sub-index from the Marketization Index Report of Chinese Provinces compiled by [Wang et al. \(2016\)](#). This index is constructed from provincial indicators such as patent application intensity, efficiency in handling patent infringement cases, and the strength of judicial enforcement. Higher values indicate stronger IPR enforcement. Provinces are classified into low- and high-IPR groups based on the median of the index.

Columns (3) and (4) of Table 8 show that firms located in provinces with weaker IPR enforcement experience significantly negative innovation responses to deep integration in export destinations, whereas firms in high-IPR provinces exhibit no statistically significant response. When we further divide provinces into quintiles based on IPR strength, Figure A4 shows a clear gradient: firms in provinces with the strongest IPR enforcement display positive innovation responses, while those in provinces with weaker IPR enforcement exhibit negative responses to deep integration in export destinations.

These findings suggest that the innovation effects of deep integration of export destinations critically depend on local institutional conditions. Strong IPR enforcement enhances firms’ ability to

appropriate the returns to innovation, allowing them to benefit from improved market access and to escape heightened competition. By contrast, in regions with weak IPR enforcement, intensified competition induced by deep integration of export destinations reduces expected innovation rents and depresses innovation incentives, leading firms to scale back innovation activities. In other words, deep integration of export destinations is more likely to promote innovation only when supported by strong local IPR institutions.

6 Conclusion

This paper examines how firms in non-member countries adjust their innovation strategies when their export destinations engage in deep integration through PTA membership. We approach this research question from the perspective of China as a third country. Using Chinese firm-level data and detailed information on PTA provisions, we construct the local exposure of an industry to the deep integration of its export destinations and find it affects firms' innovation negatively.

The Schumpeterian effect and the escape-competition effect are potential mechanisms at work through which DI affects the innovation behavior of Chinese firms. Both export quantity and price are negatively affected and are equally important channels in accounting for the trade diversion effect, a driving force of the negative innovation effect. Meanwhile, exporters export fewer products and to fewer countries. These results substantiate the fierce competition in the export destinations, which in turn discourages firms from innovating, consistent with the Schumpeterian effect. On the other hand, we find the least productive firms suffer profit losses and exhibit the greatest negative innovation response to DI. Interestingly, the negative effects are alleviated for more productive firms. The most productive firms achieve the highest profits and the greatest innovation, in line with the escape-competition effect. The escape-competition effect is stronger in sectors with higher R&D intensity. We also show that firms' innovation responses to the deep integration of export destinations vary by PTA provisions, China's PTA partnership, firms' patent types, upstreamness, and local IPR enforcement.

The results in this paper have important policy implications. The heterogeneous innovation responses across the TFP spectrum highlight the importance of strengthening domestic IPR enforcement to safeguard and promote firm-level innovation. Enhanced IPR enforcement would foster a more supportive environment for firms to innovate and develop competitive technologies, enabling them

to better navigate the intense competition in export markets undergoing deep integration through PTAs. This is particularly crucial for highly productive firms operating in R&D-intensive sectors or exporting to destinations that are China’s PTA partners. On the other hand, proactively expanding PTA networks, strengthening MFN provisions (such as IPR), and facilitating intermediate-goods trade are complementary and feasible policy strategies. This dual approach aligns economic growth with innovation-driven development, offering an effective policy pathway for sustainable economic advancement.

Our findings also have welfare implications. Innovation is an important source of economic growth. However, in response to the deep integration of export destinations, Chinese firms experienced negative innovation on average. Meanwhile, the decline of export price against unchanged import price suggests a deterioration of the terms of trade. Both seem to suggest a potential welfare loss for China. Future research could focus on developing a structural model to assess the overall third-country welfare impact, accounting for the heterogeneous effects on innovation along different dimensions such as firms’ TFP or PTA provisions.

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Appendix

A Additional Figures

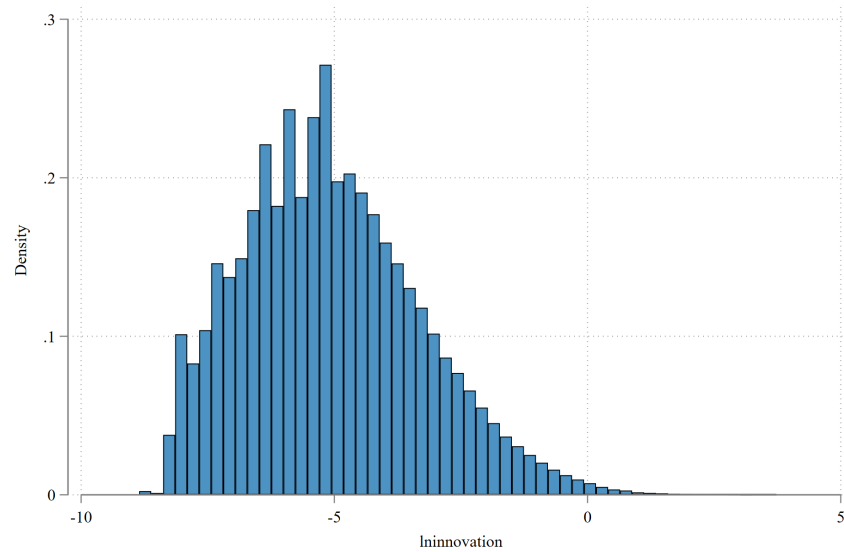


Figure A1: Innovation index (logged) at the city-industry level by [Kou and Liu \(2017\)](#).

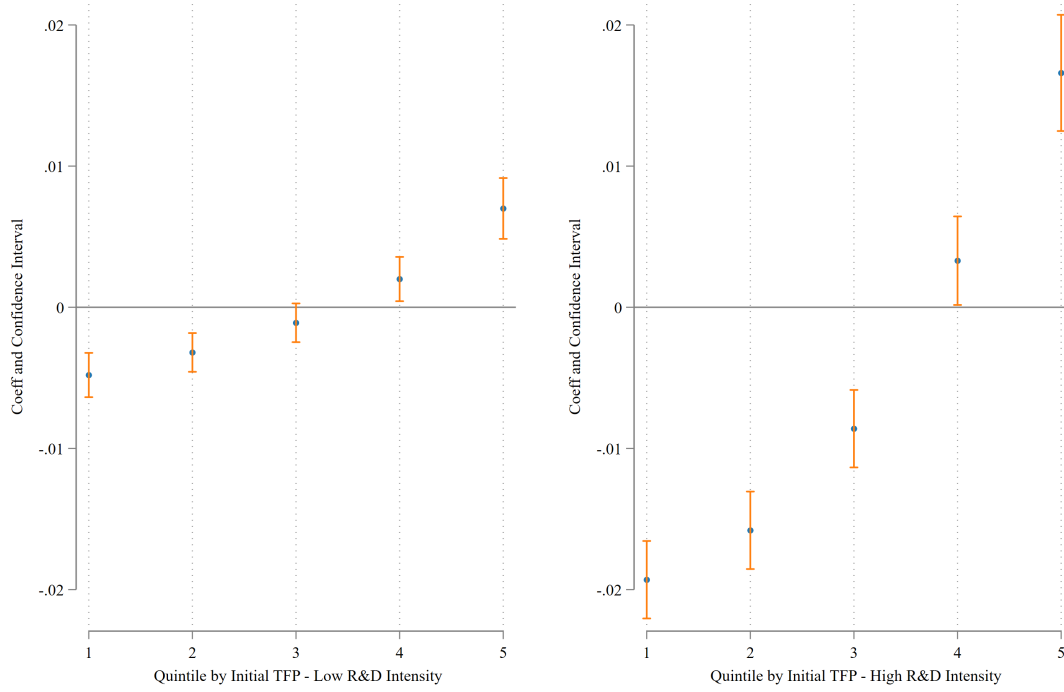


Figure A2: Heterogeneous innovation effects by TFP and R&D intensity. The plots show the coefficient α_1^g of regression (6) for each TFP quintile g and the corresponding 95% confidence interval, where g ranges from 1 to 5. The dependent variable is firm f 's invention patents. The left plot focuses on the subsample of sectors with low R&D intensity, and the right plot focuses on the subsample of sectors with high R&D intensity.

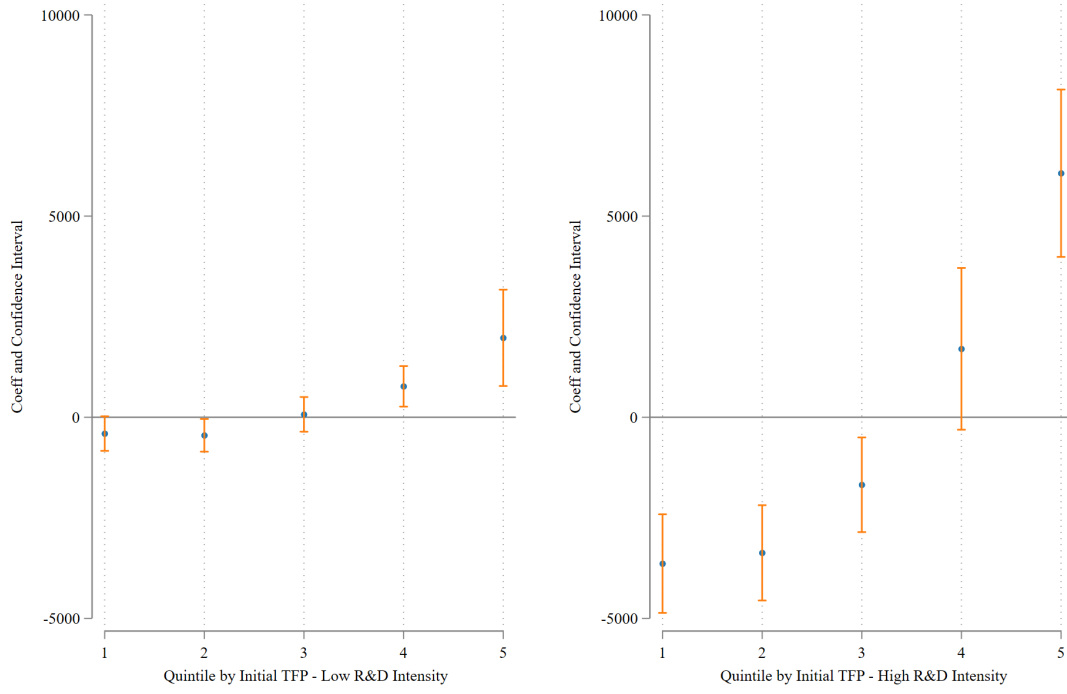


Figure A3: Heterogeneous profit effects by TFP and R&D intensity. The plots show the coefficient α_1^g of regression (6) for each TFP quintile g and the corresponding 95% confidence interval, where g ranges from 1 to 5. The dependent variable is firm f 's profit level. The left plot focuses on the subsample of sectors with low R&D intensity, and the right plot focuses on the subsample of sectors with high R&D intensity.

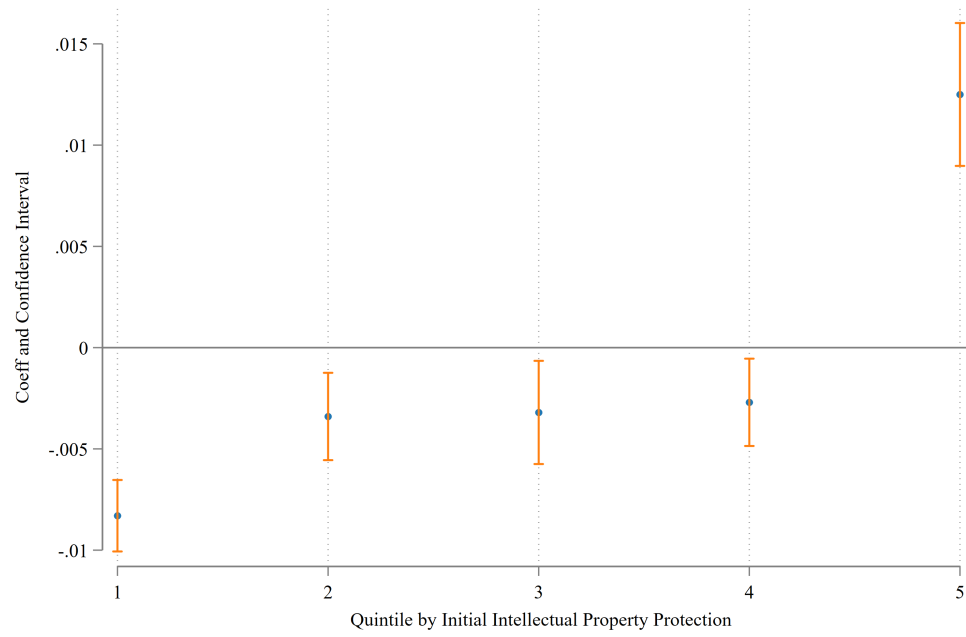


Figure A4: Heterogeneous innovation effects by firms' local IPR protection. The plots show the coefficient α_1^g of regression (6) for each IPR protection quintile g and the corresponding 95% confidence interval, where g ranges from 1 to 5. The dependent variable is firm f 's invention patents.

B Additional Tables

Table B1: Categorization of WTO+ and WTO-X provisions

WTO+	WTO-X	
Tariff industrial goods	<i>Competition policy</i>	Health
Tariff agricultural goods	<i>Investment measures</i>	Human rights
<i>Customs</i>	<i>Movement of capital</i>	Illegal immigration
Export taxes	<i>Intellectual property rights (IPR)</i>	Illicit drugs
<i>SPS measures</i>	Labor market regulation	Industrial cooperation
<i>State trading enterprises</i>	Environmental laws	Information society
<i>TBT measures</i>	Visa and asylum	Anti-corruption
Countervailing measures	Consumer protection	Mining
Anti-dumping	Data protection	Money laundering
<i>State aid</i>	Approximation of legislation	Nuclear safety
Public procurement	Agriculture	Political dialogue
<i>TRIMS measures</i>	Audiovisual	Public administration
<i>GATS</i>	Civil protection	Regional cooperation
<i>TRIPS</i>	Innovation policies	Research and technology
	Cultural cooperation	SMEs
	Economic policy dialogue	Social matters
	Education and training	Statistics
	Energy	Taxation
	Financial assistance	Terrorism

Note: The provisions in bold are the core provisions, of which the MFN provisions are italics. Non-italicized bold provisions are preferential ones.